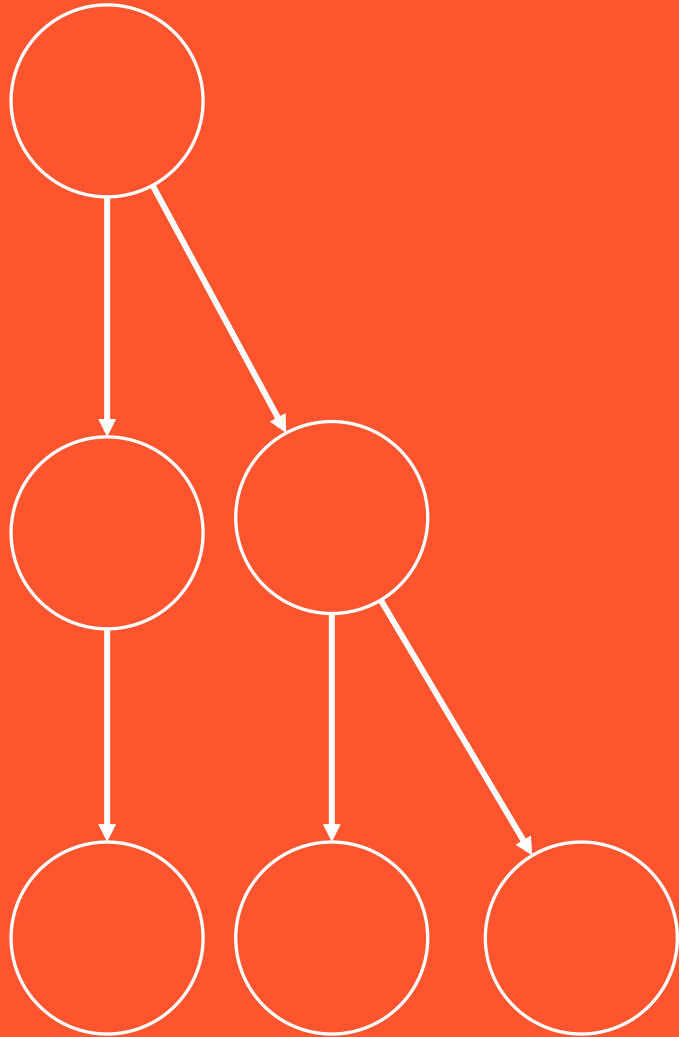




# BCOG 100

Neural Circuits, Affect, and Valence  
Lecture 4.2 — Circuits That Compute Value

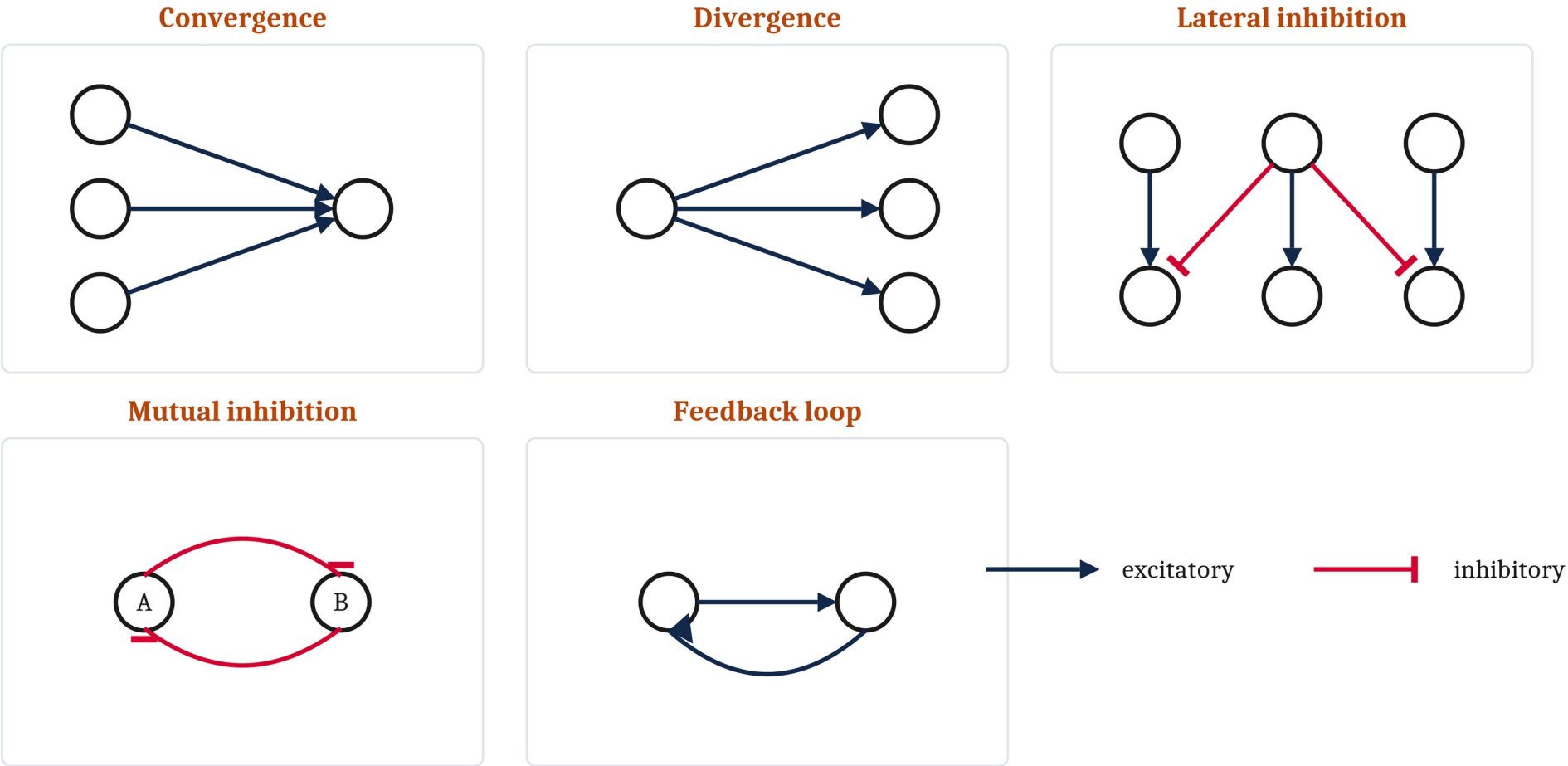


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# From a Unit to a Wiring Diagram

Connection is a matter of **wiring**, not of proximity — two cells pressed together may have nothing to do with each other. So a nervous system can be **drawn**, and the drawing reasoned about without knowing anything about the cells.

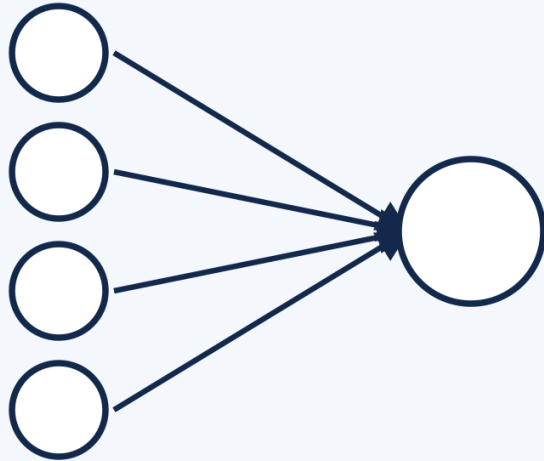
Five wiring patterns, and between them they do the work



# The Two Simplest Motifs

**Convergence** — many cells onto one — is what makes combining possible, because a cell can only weigh what actually arrives at it. **Divergence** — one cell onto many — is what makes an action a whole-body commitment rather than a local twitch.

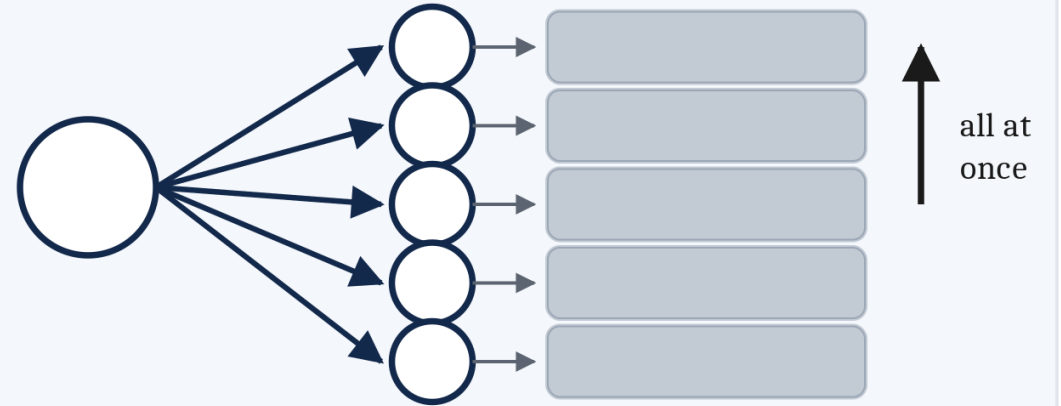
convergence



a cell can only weigh what actually arrives at it

the demand:  
**COMBINE**

divergence



one verdict, reaching muscle all over the body

the demand:  
**COMMIT**

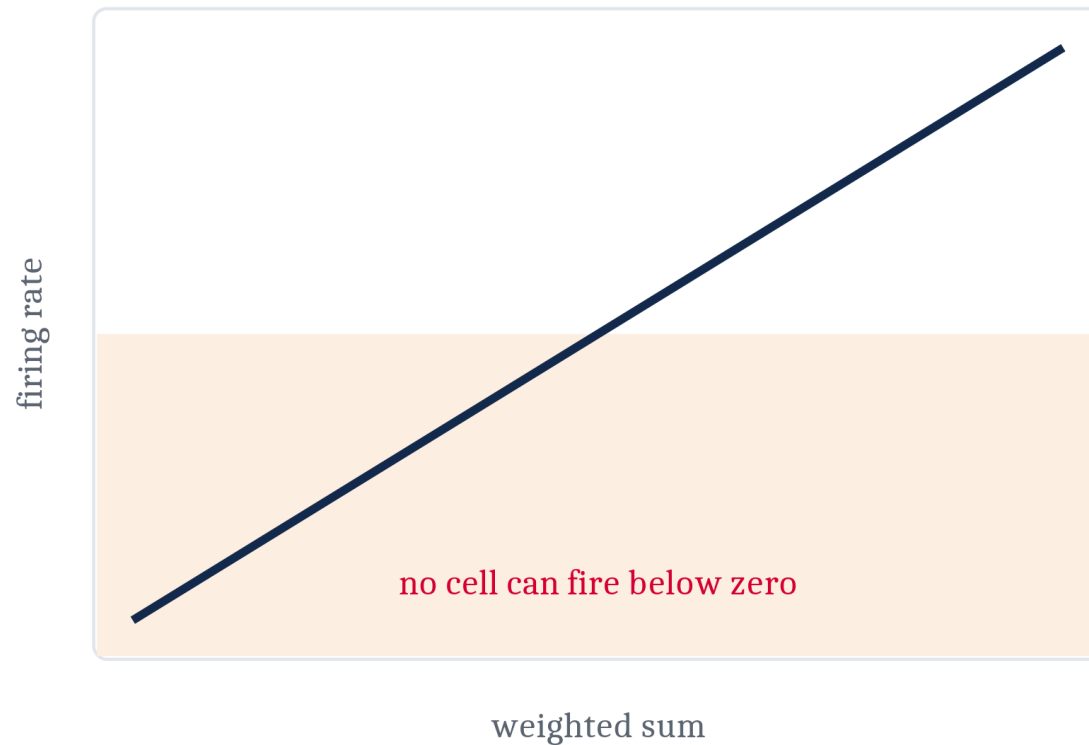
*Many onto one makes combining possible. One onto many makes an action a whole-body commitment.*

# The Floor and the Ceiling

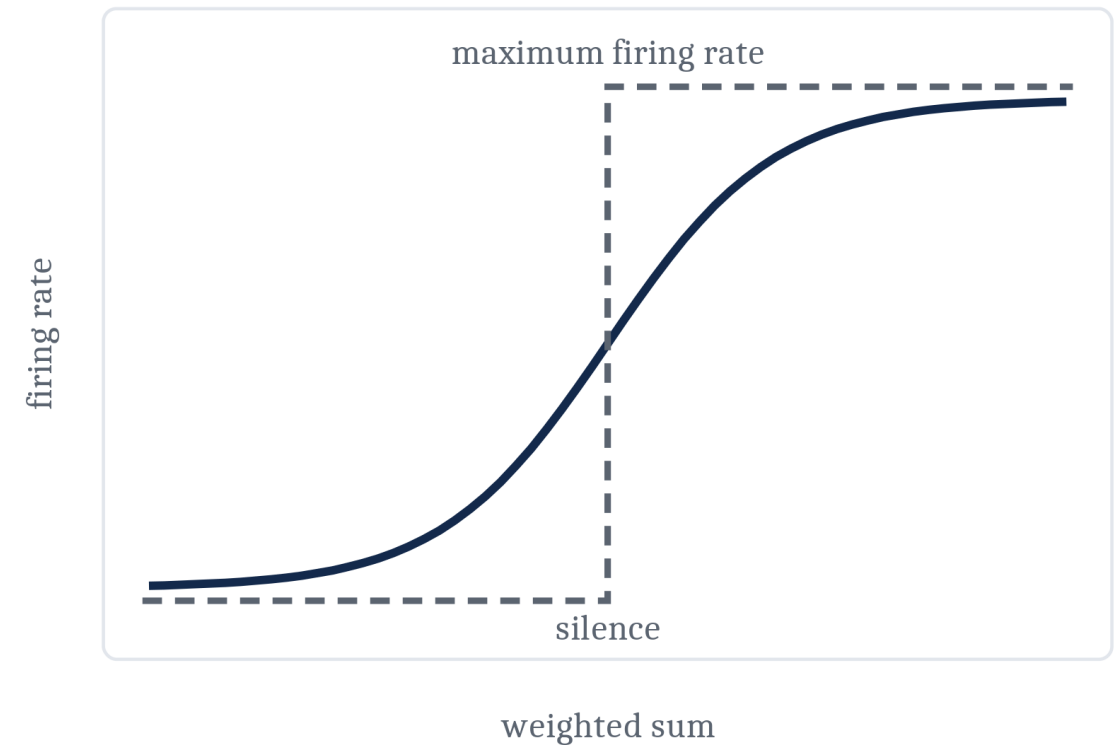
A cell cannot fire less often than never, and cannot fire faster than its refractory period allows. So the weighted sum cannot be the final answer: it passes through an **activation function** — a hard **threshold function**, or a smooth **sigmoid function**.

## Adding a floor and a ceiling

Module 3's weighted sum



with an activation function



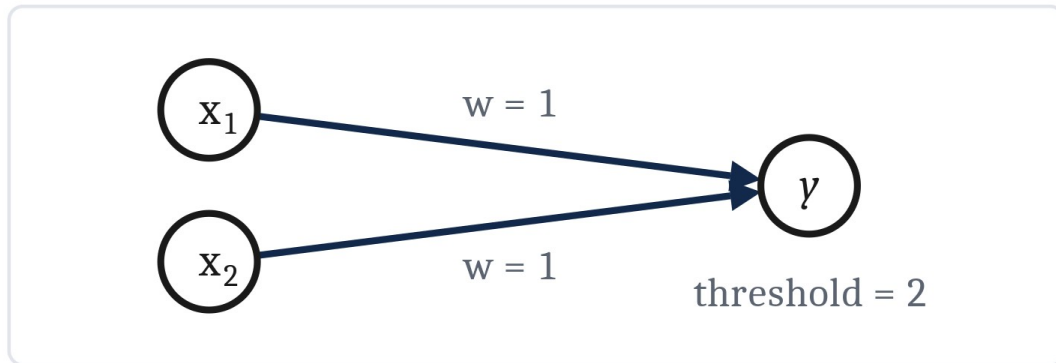
A hard threshold and a smooth S-shaped curve solve one problem two ways.

# Two Numbers Make a Logic Gate

Two excitatory inputs, each with a weight of 1. Threshold 2 and the unit computes *AND*. Threshold 1 and it computes *OR*. **Nothing about the wiring changed** — only the threshold.

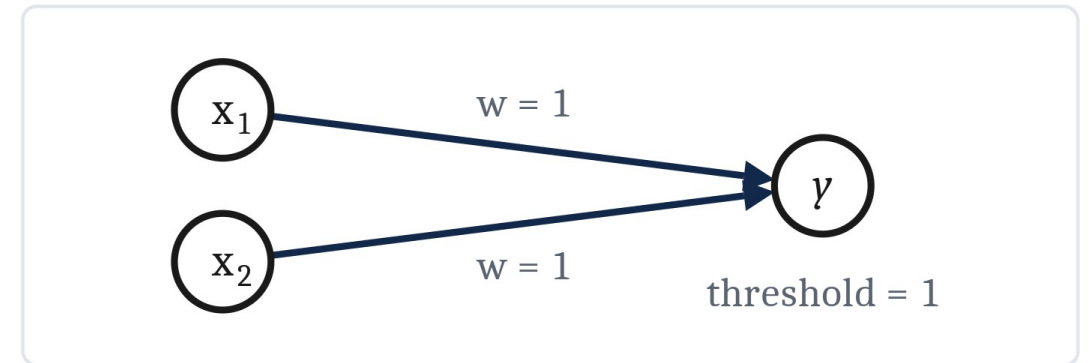
Identical wiring. One number changed.

**AND**



| $x_1$ | $x_2$ | $y$      |
|-------|-------|----------|
| 0     | 0     | 0        |
| 1     | 0     | 0        |
| 0     | 1     | 0        |
| 1     | 1     | <b>1</b> |

**OR**

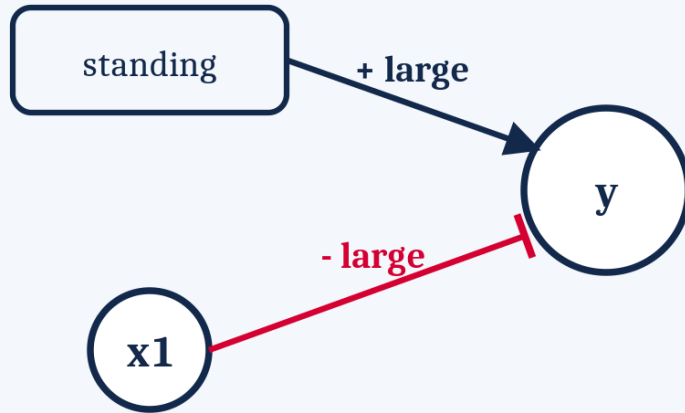


| $x_1$ | $x_2$ | $y$      |
|-------|-------|----------|
| 0     | 0     | 0        |
| 1     | 0     | <b>1</b> |
| 0     | 1     | <b>1</b> |
| 1     | 1     | <b>1</b> |

# This, and Not That

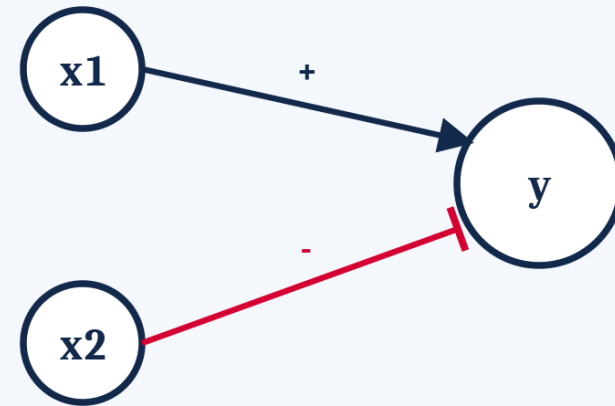
An inhibitory input is a connection with a **negative weight**. Standing excitation cancelled by one gives *NOT*; wired alongside an excitatory input it gives *AND NOT*. The unit is not choosing, and does not know what *AND* means.

## NOT



**x1 silent -> y fires**  
**x1 firing -> y silent**

## AND NOT

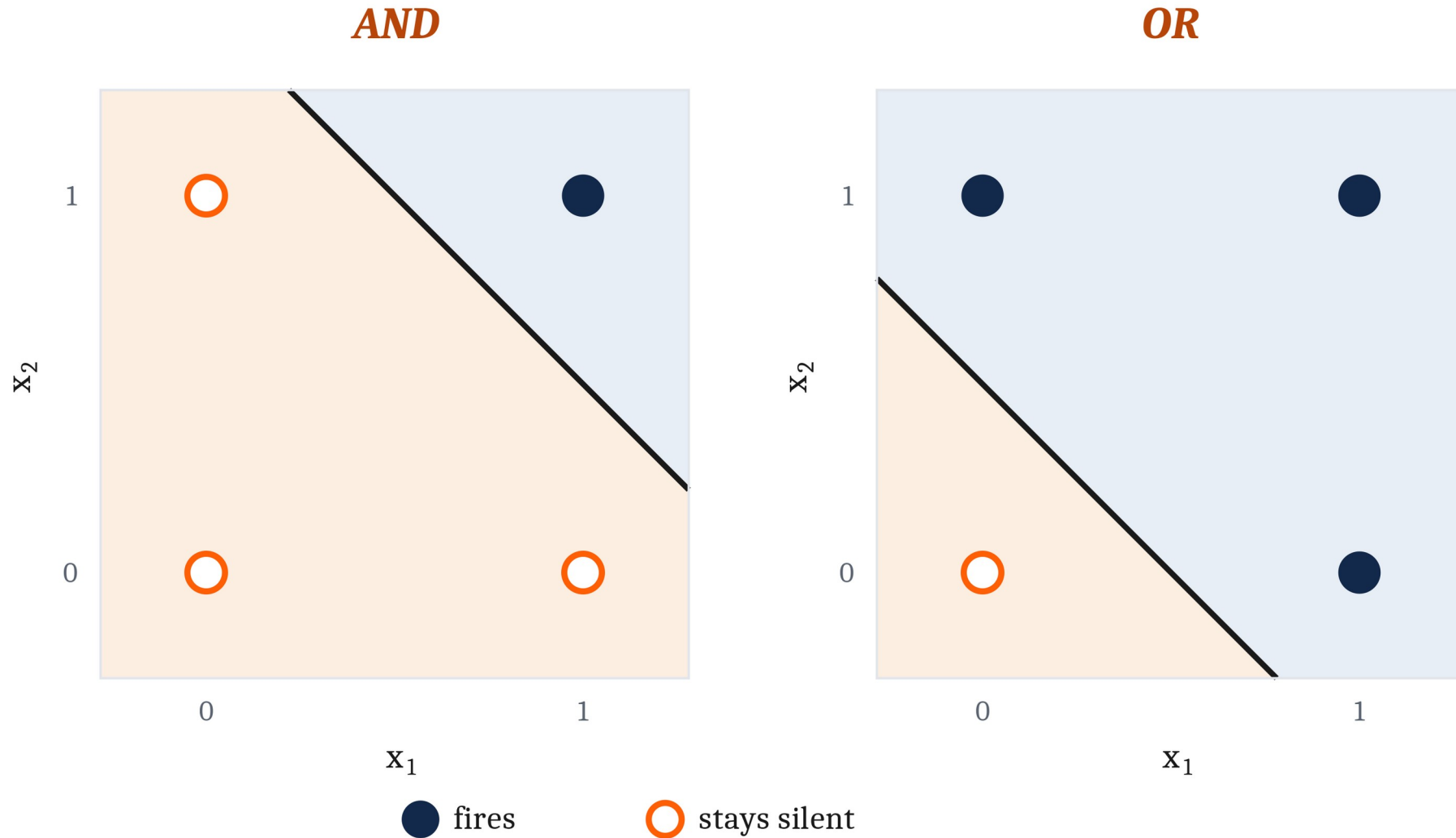


**fires when x1 is present**  
**and x2 is absent**

*The logic is in the weights and the threshold, and nowhere else. The unit is not choosing.*

# What a Unit Computes, as a Picture

Put the two inputs on a pair of axes and every combination is a point on a plane. A unit with a threshold or sigmoid



# Two Knobs That Slide the Same Line

A shift with no rotation is a change of **intercept** — the baseline firing rate. So raising the threshold and lowering the baseline do the same work. And one straight line is a limited thing.

two knobs

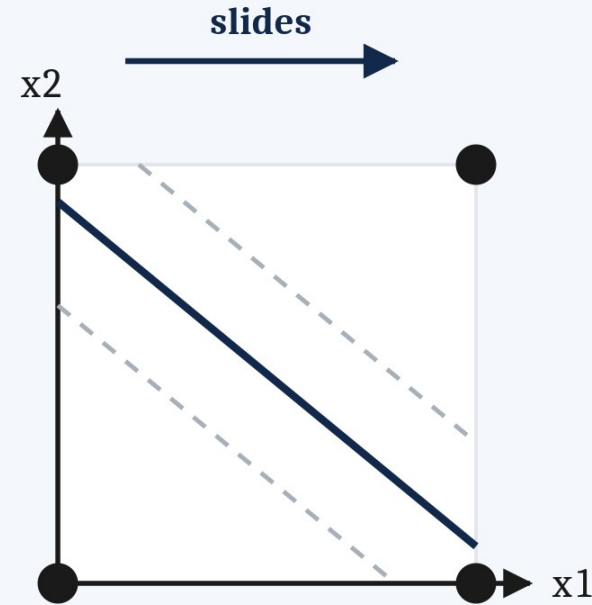


threshold  
turn it up



baseline firing rate  
turn it down

*both slide the line the same way*



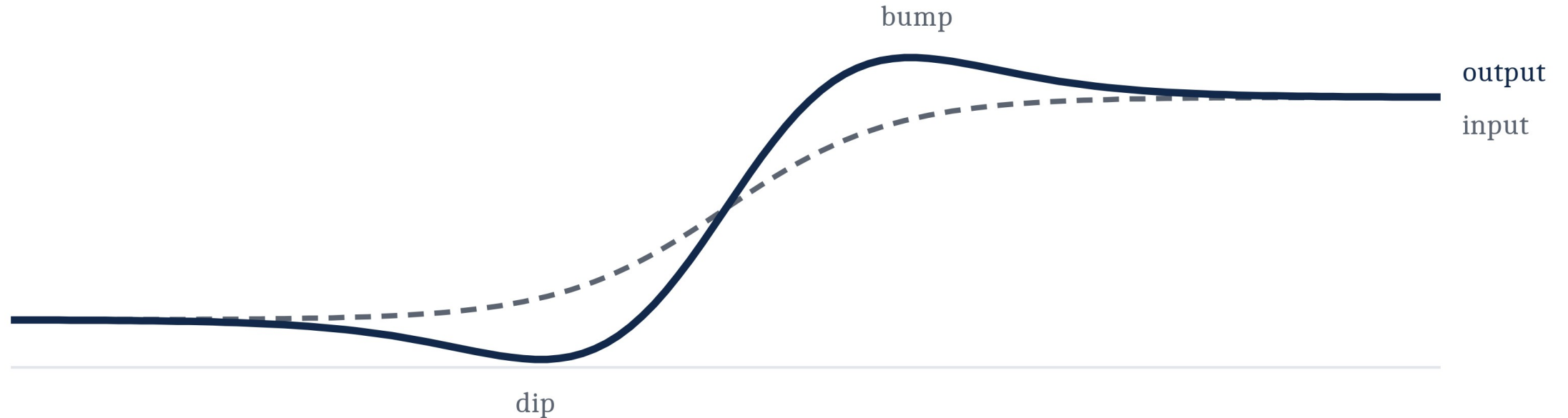
*no rotation. only position.*

*Raising the threshold and lowering the baseline do the same work.*

# Lateral Inhibition: Reporting the Difference

A row of units, each inhibiting its neighbors in proportion to its own activity. The one that is slightly ahead inhibits harder than it is inhibited, so its lead grows: **a shallow step in, a steep step out.**

**What lateral inhibition does to a gentle edge**

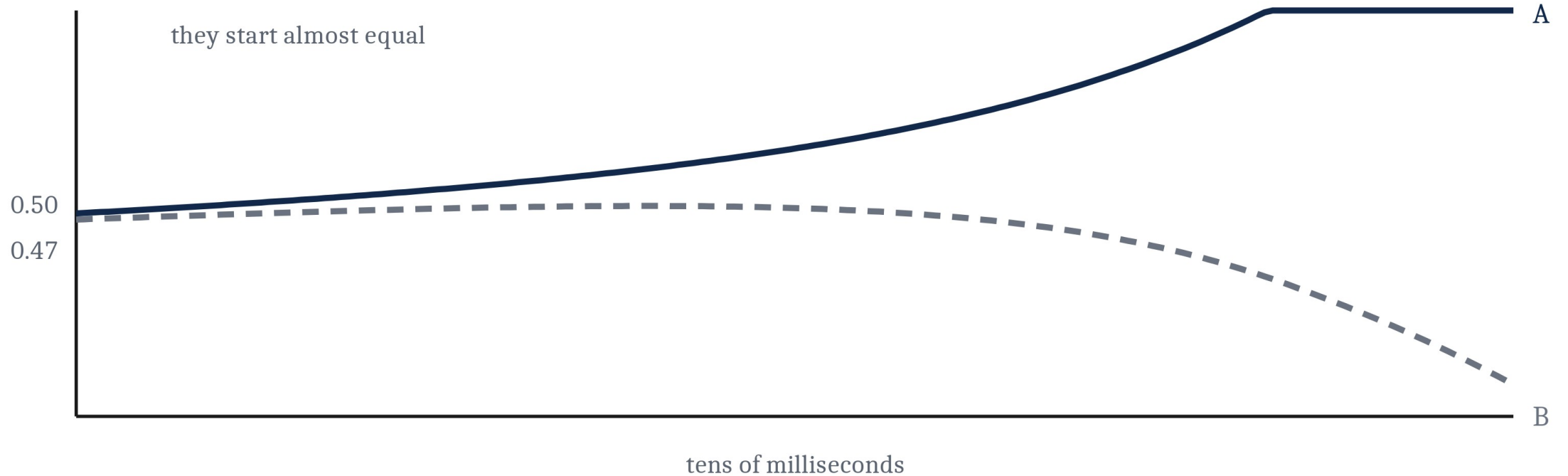


Same overall levels. A much steeper transition.

# Mutual Inhibition, and Choosing Exactly One

Two units, each inhibiting the other. From a near tie the lead runs away, and within a few tens of milliseconds one is firing and the other is silent. **Mutual inhibition** forces **winner-take-all**.

A near tie, resolved



The circuit does not average a near tie. It resolves one.

# A State Can Be a Circuit That Is Still Running

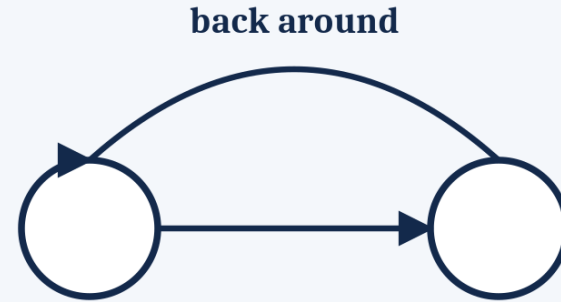
Everything so far ran one way — **feedforward**. A **feedback** or **recurrent connection** runs a unit's output back toward its own input. The stimulus lasts a moment; the activity lasts as long as the loop runs.

**feedforward**



input to unit to output, one direction

**recurrent**



every connection in the loop still runs one way

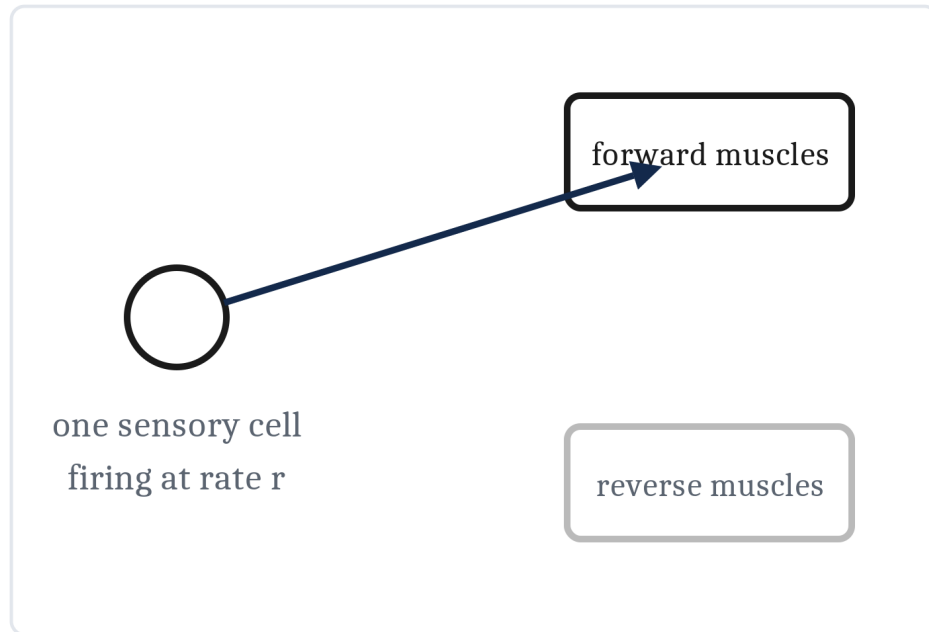


# Where Valence Lives

A sensory neuron carries a rate, and nothing else — a food cell and a toxin cell can be identical in every respect a downstream cell could measure. What makes one a good-detector is **what it is wired to**.

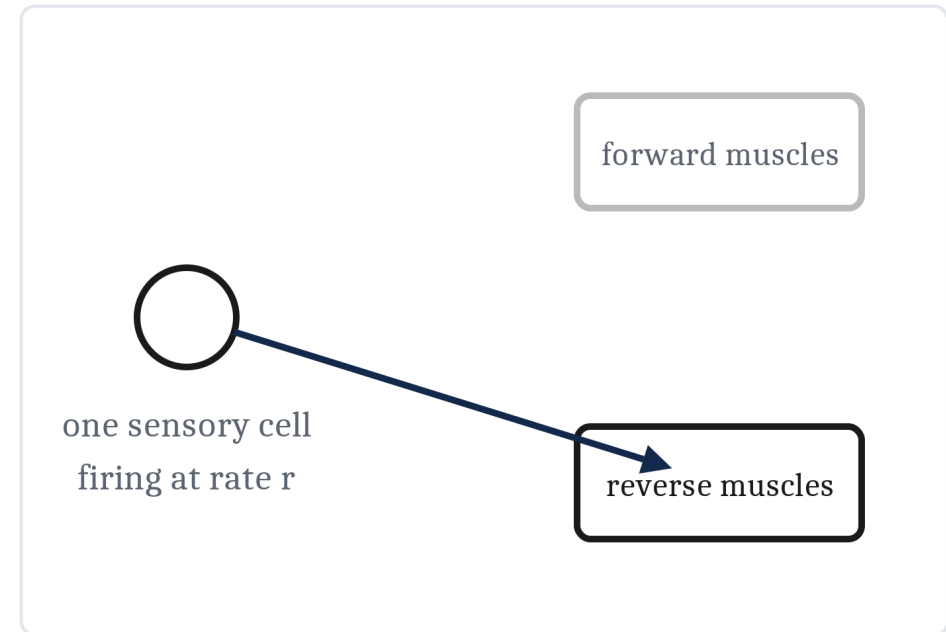
## The same cell, wired two ways

### Wired one way



the animal moves toward it

### Wired the other



the animal moves away from it

Nothing about the cell or its rate differs. Only what it is wired to.

# Why a Space, and Not a List

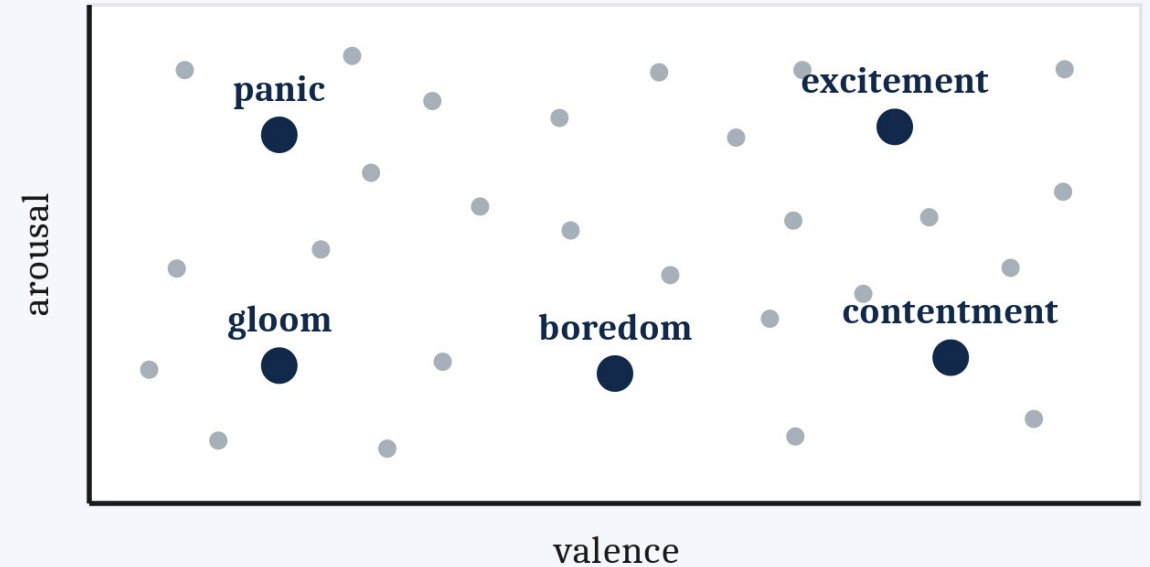
Algorithmically, an affective state is **two numbers** — how good or bad things are, and how activated the animal is — and what it does next is a function of its position. A list holds only what someone thought to name.

## a list

- fear
- anger
- joy
- sadness
- disgust
- surprise

written in advance, and holding only  
what someone thought to name

## a space

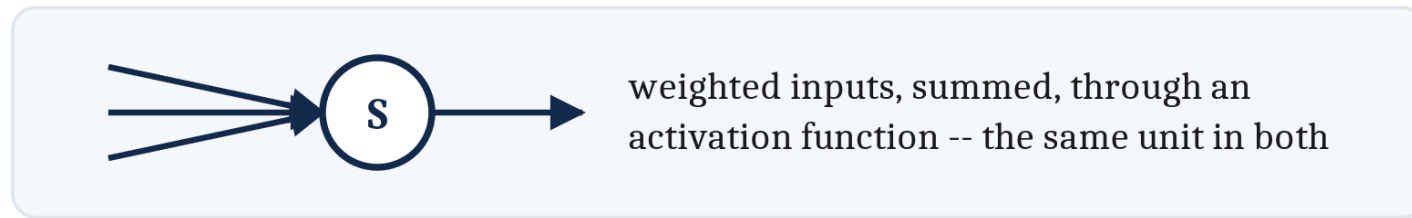


every position is a possible state, including the ones with no word

*And two numbers describe a worm and a person in the same terms.*

# The Same Motifs in Silicon

An artificial network is also built from units that take weighted inputs, sum them, and pass the total through an activation function — and it is wired into the same patterns. One diagram serves both.



**convergence and divergence**



every layer of a network is made of many-onto-one and one-onto-many

**mutual inhibition**



wherever a network has to settle on one output among several

**a recurrent loop**

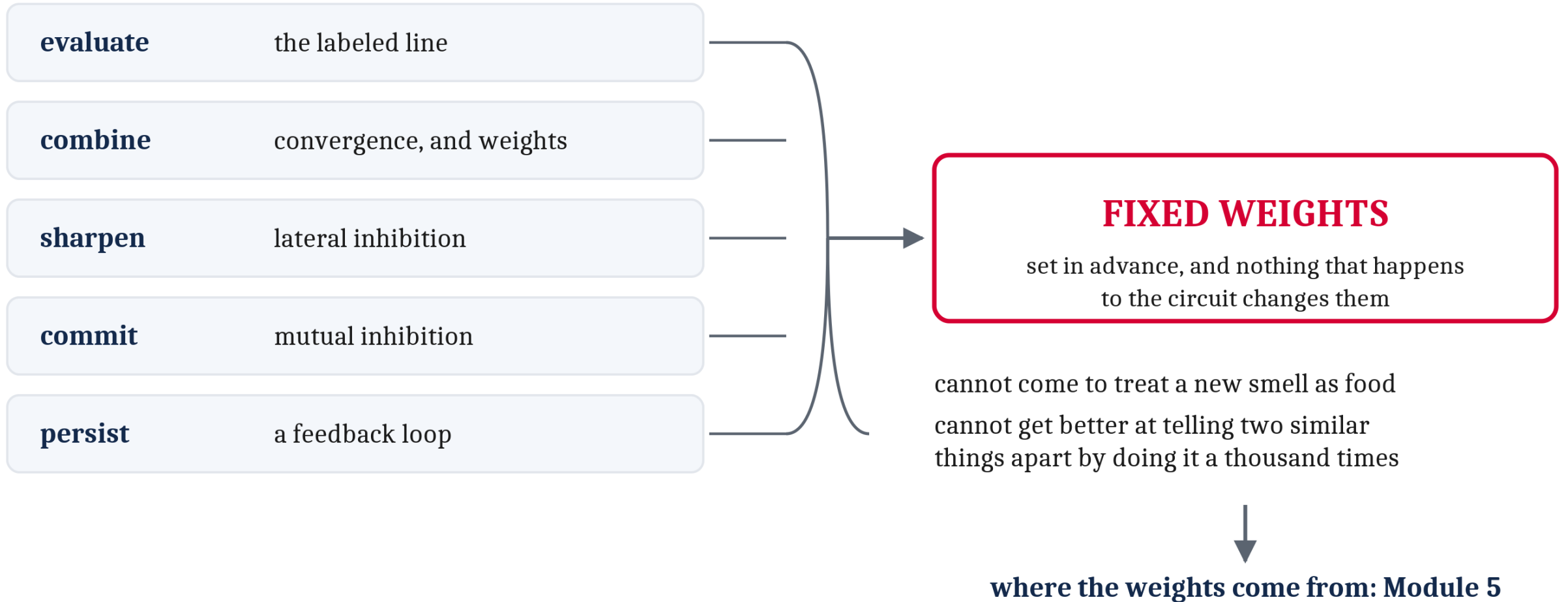


how a network holds a state across time

*One diagram for both -- Module 1's multiple realizability, at the scale of the circuit.*

# What a Fixed Circuit Cannot Do

Every demand raised in the first lecture now has a wiring pattern that meets it. And every one of those arrangements has **fixed weights** — set in advance, unchanged by anything that happens to the circuit.



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| 14    | The motifs, matched to their uses in artificial networks | fig-motifs-in-silicon.png | CC BY 4.0 | Jon Willits — drawn for this lecture · own work         |
| 15    | Every demand met, and the one limit they share           | fig-fixed-weights.png     | CC BY 4.0 | Jon Willits — drawn for this lecture · own work         |